

DFT – Molecular Dynamics Simulation

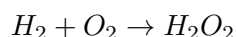
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1. Introduction

The aim of this practice is to study the influence of the initial velocity on the dynamical collision between two molecules through *ab initio* molecular dynamics simulations using SIESTA.

The analysis focuses on how variations in the initial kinetic energy affect the outcome of the collision, leading to weak interactions, elastic collisions, or a possible reaction.

In this simulation, two oxygen atoms and two hydrogen atoms are considered, reproducing the following reaction:



2. Methodology

The SIESTA code was used to perform *ab initio* molecular dynamics simulations.

To carry out this study, three different cases were considered in which only the velocity component v_x of the hydrogen molecule (H_2) was modified.

All other parameters were kept unchanged from the initial conditions.

The three studied cases are:

- Case A: original velocity
- Case B: reduced velocity ($v_x=-0.3$)
- Case C: increased velocity ($v_x=-0.8$)

Case A was kept unchanged, as it serves as a reference (control case) to ensure that the simulation setup is correct.

For each simulation, the following items were analysed:

- Atomic trajectory from the .ANI file
- Temporal evolution of a relevant interatomic distance
- Temperature evolution
- Potential and total energy of the system from the .MDE file

In addition, the number of MD steps was increased in order to ensure that the collision between the molecules occurs within the simulated time window. Specifically, the MD of steps was increased from 20 to 100.

3. Results

3.1. Trajectory

Case A

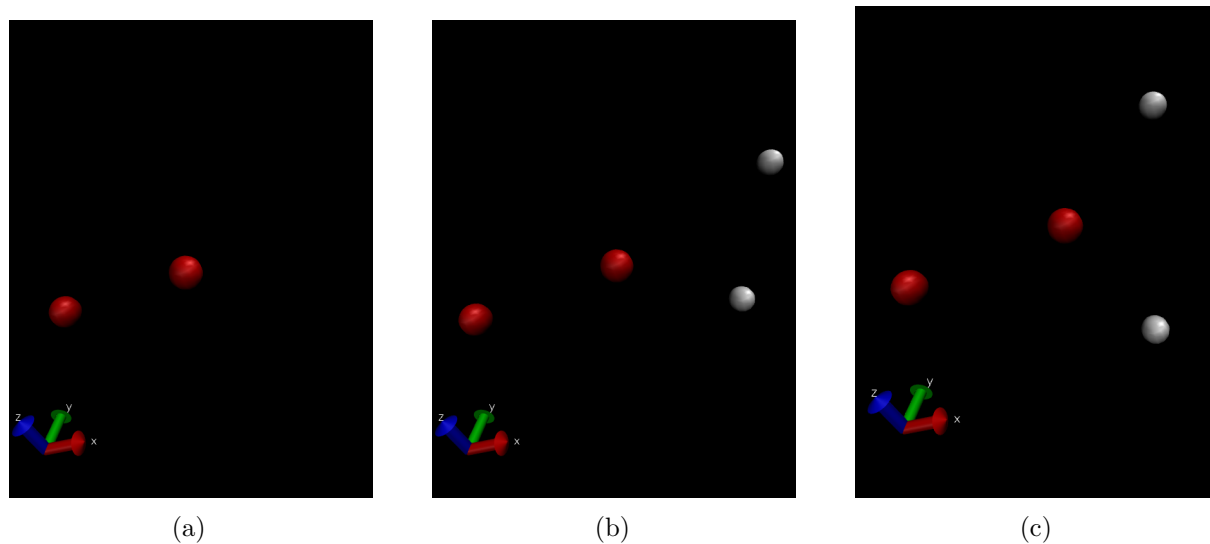


Figure 1: Trajectory of case A.

As can be seen in Figure 1, the hydrogen molecule approaches the oxygen molecule; however, the simulation appears to stop before any significant interaction occurs.

The simulation finishes without bond formation or chemical reaction. No effective collision is observed within the simulated time window, as the molecules remain separated in the final configuration.

Case B

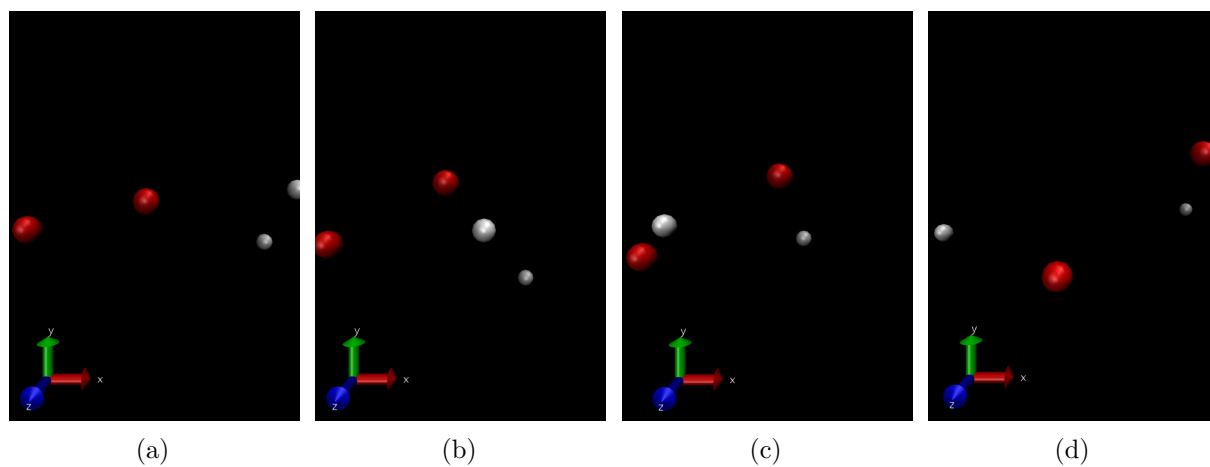


Figure 2: Trajectory of case B.

In Figure 2, the trajectory corresponding to the lowest initial velocity is shown. The H₂ molecule approaches the O₂ molecule gradually, reaching a minimum intermolecular distance, which

corresponds to the point of strongest interaction between them.

However, no effective collision or chemical reaction is observed, as the initial kinetic energy is not sufficient to overcome the potential energy barriers. After the interaction, the molecules separate again.

In this simulation, the process had to be manually stopped, as the hydrogen atoms began to orbit around the oxygen molecule. This behavior may be related to the system exploring configurations of lower energy, leading to a transient, quasi-bound state without the formation of a stable chemical bond.

Case C

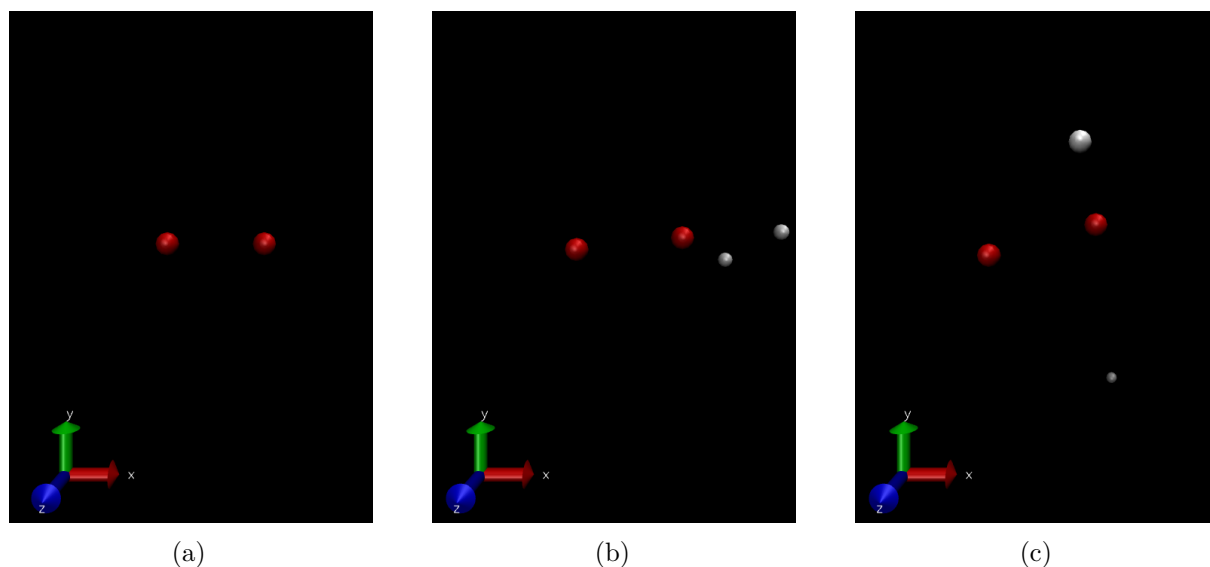


Figure 3: Trajectory of case C.

As can be seen in Figure 3, the trajectory corresponding to the highest initial velocity is shown.

As expected, the hydrogen approaches the oxygen molecule very rapidly. In this case, the kinetic energy is high, resulting in a very short interaction between the molecules. Consequently, no significant interaction occurs, and the system undergoes an elastic collision.

After the collision, the molecules separate again, with the hydrogen molecule moving away from the oxygen without any bond formation or chemical reaction.

3.2. Distance evolution

In order to obtain this plot, a distance measurement between one hydrogen atom and one oxygen atom was defined using a tool from VMD. The distance between these atoms was measured along the step simulations and plotted for the three studied cases. This temporal evolution is shown in Figure 4.

In case A, the distance decreases gradually but does not reach a value indicative of a collision. This confirms the absence of a significant interaction between the molecules.

In case B, the distance also decreases, although the approach is slower than in case A due to the lower initial velocity. The distance reaches a minimum corresponding to the closest approach

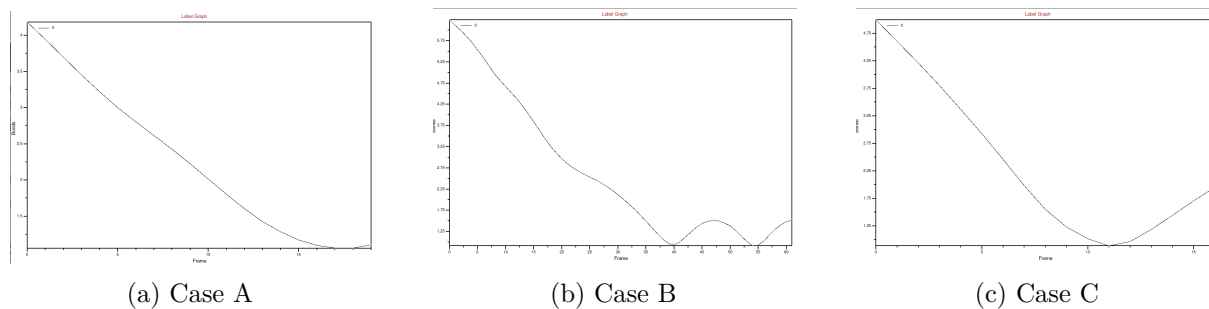


Figure 4: Distance evolution

between the molecules. However, after this point, the distance increases again.

In case C, the distance decreases rapidly, reaching a minimum in a short time, and then increases again immediately.

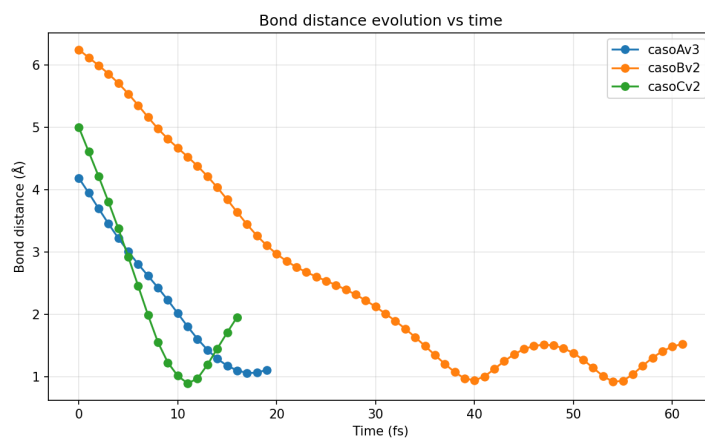


Figure 5: Time evolution of the interatomic distance for the three cases.

3.3. Energy

Figure 6 shows the evolution of the Kohn–Sham (potential) energy and the total energy as a function of time for the three studied cases.

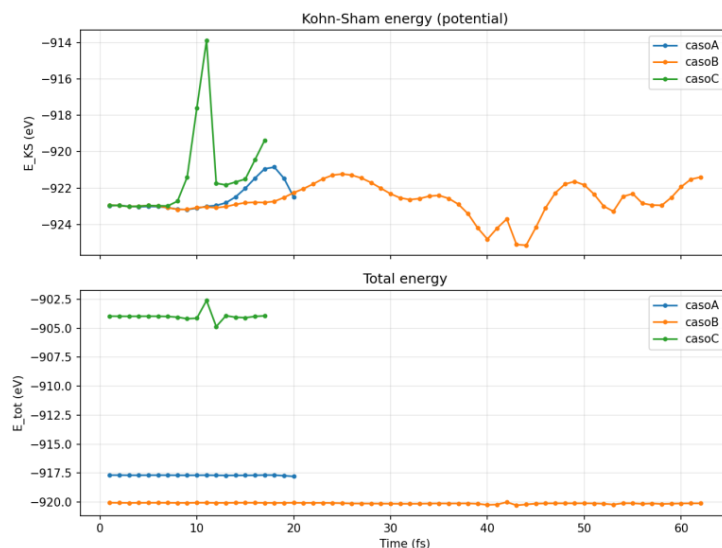


Figure 6: Time evolution of the energy.

Figure 6 shows the evolution of the Kohn–Sham (potential) energy and the total energy as a function of time for the three studied cases.

In the upper panel, the Kohn–Sham energy exhibits noticeable variations during the simulation, particularly around the time of closest approach between the molecules. In case C (green), a sharp increase in the energy is observed, indicating a strong but very brief interaction. In case B (orange), the energy changes more gradually, reflecting a longer interaction time. In case A (blue), the variation is smoother and less pronounced.

In the lower panel, the total energy remains approximately constant for all cases, with only small fluctuations. This confirms that energy is conserved during the simulations.

The total energy remains approximately constant within each simulation; however, different cases exhibit different absolute values due to their distinct initial conditions. As can be observed, the energy profiles are nearly flat, with small fluctuations that can be attributed to expected numerical errors. Between different cases, each curve lies at a different energy level (approximately 917, 920, and 905 eV). These differences arise from the variation in kinetic energy, which depends on the initial velocity assigned to each system.

3.4. Temperature

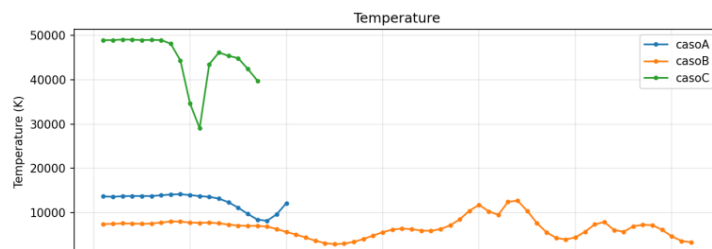


Figure 7: Time evolution of the temperature.

Figure 7 shows the evolution of the temperature as a function of time for the three studied cases.

The temperature is related to the kinetic energy of the system. In case A, it remains almost constant, indicating weak interaction. In case B, it shows noticeable fluctuations due to the collision. In case C, the temperature is initially high and decreases rapidly, reflecting a fast interaction.